



Heuristic assumptions

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Abstract

Interpreting any decision requires making auxiliary assumptions regarding how the decision makers viewed their options and related them to their lives. Theories constrain those assumptions. The more general the theory, the fewer constraints it imposes and the more assumptions must be made in any application. Like the rational actor models that they challenged, Daniel Kahneman and Amos Tversky's heuristics-and-biases and prospect theory research programs were general theories, with broad application. One of the many marvels of their landmark studies is that they rarely test their auxiliary assumptions. Rather, readers were trusted to agree about how people interpret the tasks (e.g., select anchors in studies of that heuristic). Subsequent studies have often accepted those interpretations in order to examine boundary conditions (e.g., extreme anchors). Applying the theories to naturally occurring tasks requires making additional auxiliary assumptions. This article illustrates three ways to evaluate those assumptions: direct assessment, systematic manipulation, and archival analysis. It concludes with proposals for loosely coordinated evaluation of shared and contested assumptions.

Keywords Daniel Kahneman · Amos Tversky · Judgment · Heuristics · Prospect theory · Auxiliary assumptions

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1 Introduction

Any theory with empirical content requires auxiliary assumptions that link its constructs to observable events. The more general the theory, the more assumptions are needed. Toward one extreme, lie botanical theories about organisms that survive only in narrow habitats, such as orchids that require well-drained acidic soil, a narrow temperature band, morning mists, and rare pollinators. Toward the other extreme, lies utility theory, whose constructs can apply to any decision, as long as the options, outcome utilities, and outcome probabilities can be specified precisely enough to allow predictions.

With great power comes great responsibility. A theory is empty without constraints on its assumptions. With enough ingenuity, one can imagine some set of options, utilities, and probabilities that allow construing any choice as following utility theory. One constraint on these assumptions is stable preferences (Becker, 1976). People must have consistent utilities across a set of decisions specified by auxiliary assumptions. It is researchers' duty to delineate and test those limits. An example is the work of Kip Viscusi and others in determining the bounds on value-of-life utilities (Viscusi, 2018).

Unless these assumptions are explicitly defined and tested, there are no constraints on a theory, making it less than a theory (e.g., St Quinton & Trafimow, 2022). The preference reversals demonstrated by Sarah Lichtenstein and Paul Slovic identified an influential failure of an auxiliary assumption of utility theory. The extensionality assumption holds that preferences should be the same, however they are elicited—as long as the options, utilities, and probabilities remain the same. Lichtenstein and Slovic (1971, 1973) created gambles that elicited different preferences when the question was posed in different ways (willingness to pay vs. preference for playing). That finding constrained the set of potentially valid auxiliary assumptions, prompting research into what those constraints were, notably framed by David Grether and Charles Plott (1979).

2 Boundary conditions

A marvel of Danny Kahneman and Amos Tversky's epic research programs, heuristics and biases (Tversky & Kahneman, 1974) and prospect theory (Kahneman & Tversky, 1979), is that their landmark demonstration studies rarely tested their auxiliary assumptions. Rather, those assumptions were accepted on the basis of their face validity. Readers more or less agreed intuitively, for example, about which examples people would retrieve from memory, when relying on the availability heuristic, and what their reference point would be, when evaluating prospects. Some of the success of these demonstrations reflects their captivating tasks (e.g., the Linda problem)¹. However, their staying power reflects the strength of their auxiliary assumptions.

¹ The fictional Linda is described in a way that makes it seem as though she is more likely to be an active feminist *and* bank teller than just a bank teller, violating the conjunction rule (Tversky & Kahneman, 1983).

One source of that strength was extensive pretesting of research materials. My first data-collection task, as a graduate research assistant, was administering an early version of the survey asking questions like, “Is k more common as the first or third letter of four-letter English words?”² Once Amos and Danny were confident that people understood the questions and answers of this strange task, they were ready for proper data collection,³ mostly with paid volunteers recruited through the Oregon Research Institute (the precursor to Decision Research). Their ability to refine tasks reflects the eye for nuance that psychologists acquire through their apprenticeship.

A second source was grounding in psychological research that supported their assumptions. The tasks used proven methods, like two-alternative forced-choice questions (e.g., which is more common?), a research staple when trying to hide researchers’ expectations. Empirical assumptions were often supported by previous studies. For example, memory researchers had found that people seem to automatically keep track of how often they have observed events, making reliance on memory a plausible heuristic (Hasher & Zacks, 1984). Sensitivity to change is well-documented (Helson, 1964), supporting the central role of reference points in prospect theory.

A third source of strength was using tasks repeatedly, in studies testing how well results replicated with modest variations (e.g., different participants, incentives for accuracy, time to think). Each test provides evidence regarding the auxiliary assumptions that it tests. However, such gradual probing of boundary conditions can leave many assumptions untested. The current wave of multi-site replications has often revealed surprising variability in responses to a single task, reflecting some unrecognized auxiliary assumptions (e.g., Camerer et al., 2018).

The following sections illustrate three approaches to evaluating auxiliary assumptions: direct assessment, systematic manipulation, and archival analysis. The examples are drawn from personal research experiences inspired by Danny and Amos’ work. They are framed as tests of *construct validity*, evaluating a web of inferences regarding theories, measures, and auxiliary assumptions, each of which is affirmed when expected results are observed, each of which is weakened when expectations are not met (Cronbach & Meehl, 1955).

3 Direct assessment of availability

As a graduate student, I had the great good fortune to sit in on the seminar where the evolving heuristics-and-biases research program was aired (Fischhoff, 2024). Like other observers, my friend Ruth Beyth-Marom and I spent much time discussing the research. One issue was how could one tell whether people were actually producing examples, in studies eliciting frequency judgments, as part of employing the availability heuristic. As mentioned, Bousfield and Sedgewick (1944) found that

² My special qualification was access to English-speaking classes at Hebrew University of Jerusalem programs for foreign students. My first-ever data-collection task was with Samuel Komorita at Wayne State University, where I helped with a prisoners’ dilemma study, in what now would be called behavioral game theory. I recommend this remembrance (Parks, 2007).

³ Which appeared as Study 3 in Tversky and Kahneman (1973).

how quickly people produce examples in a brief period of time predicts how many examples they could produce if they worked longer.⁴ That result was consistent with believing that there are more words with *k* as first letter than as third letter, as examples of the former come to mind more quickly, even though there are actually more of the latter.⁵

However, a competing theory of frequency judgments holds that, rather than counting individual instances (or *tokens*) of a class in memory, people assess the strength of their memory trace for that class (or *type*) (Jonides & Naveh-Benjamin, 1987). An auxiliary assumption of the availability heuristic is that people judge frequency by tokens, rather than types.⁶ As a way to test this assumption, Ruth observed that there are a few easily recalled examples of countries beginning with the Hebrew letter *yud* (roughly *y*), but no more, while there are many moderately available examples of countries beginning with the letter *kuf* (roughly *k*). If people rely on tokens, they should overestimate the relative frequency of *yud* countries when they work for a few seconds, but not when they work longer and realize that they have exhausted the *yud* countries, while *kuf* countries kept coming. However, if people rely on the trace strength of each type, time should not make much difference. We gave people either five seconds or two minutes to think about the question. We found that, with more time, their relative frequency estimates decreased for *yud* countries and increased for *kuf* countries, consistent with relying on tokens, rather than traces, and with the availability heuristic (Beyth-Marom & Fischhoff, 1977).

Not long afterward, Paul Slovic, Sarah Lichtenstein, and I began studying risk perceptions.⁷ Availability was a natural hypothesis for why people seemingly pay disproportionate attention to some risks, while neglecting others. However, interpreting responses to risks requires auxiliary assumptions regarding what people see, what they hear, how they evaluate their sources, and how they choose to respond, among other things. When people seemingly overreact to a risk, its occurrences might be overreported (e.g., crimes), hence reflect availability. However, other explanations are possible. They might see an opportunity for action when public attention focuses on a risk. They might see a need for action if a small risk suddenly increases. They might be unable to block the risk out (e.g., terror).

Our first set of studies (Lichtenstein et al., 1978) used a multi-method approach to elicit judgments of how many people die annually from 41 familiar causes. In order to interpret those judgments in terms of availability, we assessed observation

⁴ The study (which I sometimes use in class) is worth reading, as a window into early research methods, including introduction of the kymograph (using a telegraph key for real-time data collection) and records of the examples that some subjects produced. Some examples were repeated; others did not belong to the specified category, deviations that subjects might or might not have realized upon further reflection and might reflect actual memory search.

⁵ Things might be different for crossword puzzle solvers and Scrabble players.

⁶ These studies typically use “availability” for what is in memory and “accessibility” for how easily it is extracted.

⁷ Hammond (1966) argued that *perceptions* are sensory responses, whereas *judgments* are their translation into inferences. Thus, judgments based on the same perceptions can differ, depending on the task (e.g., allotting five seconds or two minutes). A somewhat analogous terminological distinction is between *hazards* and *risks*, with the former referring to the sources potentially creating the latter.

frequency in two ways. One was having people rate their direct and indirect exposure to each cause as a source of death and suffering. The second was counting the number of deaths reported in the *Eugene Register-Guard* and *New Bedford Standard Times* (Combs & Slovic, 1979).⁸ Both approaches found that, relative to statistical frequency, greater exposure to events was associated with higher frequency estimates. Other tests evaluated the auxiliary assumptions needed to interpret the numerical estimates. They included comparing judgments of relative frequency ratios with the ratios derived from direct frequency estimates, as well as looking at response patterns for analogous estimates of the frequencies of words and of occupations. All those tests made for a very long paper. For me, it was a master class in experimental design, led by Sarah and Paul. To the best of my knowledge, these tests have never been repeated in full.

4 Systematic manipulation of reference points

A concurrent line of research, also with Paul and Sarah, examined preferences for policies affecting risks to human life. Arguably, every life should be equal. If so, then people should have utility functions that are linear in number of lives and prefer policies with the highest expected number of lives saved (or lowest expected number of lives lost). At the time, a widely discussed observation was that people seem to pay more attention to lives lost at one time than to the same number lost at one at a time (Schelling, 1968). Many studies have investigated the sources of this seeming preference, as well as potential remedies for when it seems inappropriate (Cohen et al., 2015; Slovic, 2020). A contrary observation, at the time, in the early days of risk analysis, was that people seem to demand greater benefit from technologies that can kill many people at once (e.g., nuclear power) than from technologies that exact a similar toll one death at a time (e.g., motor vehicle accidents) (Fischhoff et al., 1978; Starr, 1969).^{9,10}

One way to probe intuitive utility functions is by posing simple gambles, such as asking people to imagine advising “the civil defense committee in a large metropolitan area” about the choice between an emergency response plan where 50 lives will be lost or one where there is an equal chance of losing 40 lives or 60 lives. For people employing an expected utility decision rule, preferring the former program (50 lives lost) would reflect a utility function that puts less value on each additional life.

From a prospect theory perspective, the choice should depend on the reference point. Was it (a) 0, the current situation, with no lives lost; (b) 40, the number of lives lost whichever program is selected; or (c) 50, the certain option?¹¹ Choosing the reference point is part of the editing step in prospect theory, preceding the evaluation

⁸ A much more arduous task back then, without current technology.

⁹ This study has had a complete replication (Glassman-Fox & Weber, 2016).

¹⁰ Subsequent research found that a critical difference is that people place less trust in expected fatality estimates for technologies with catastrophic potential (Slovic et al., 1984). Thus, their behavior reflects uncertainty (or ambiguity) aversion, rather than risk aversion.

¹¹ It could also be 60, the worst-case option. For reasons that I can no longer reconstruct, I did not study it.

step, calculating expected value relative to the reference point. According to prospect theory, frames (a) and (b) lead to preferring the gamble, frame (c) to preferring the certain outcome.

Prospect theory suggests the status quo as the default auxiliary assumption for the reference point. In this case, that would be 0 lives lost [frame (a)]. However, it is not hard to write text framing the decision in the other two ways. That was one strategy that I pursued in a series of seven experiments examining the auxiliary assumption of what the reference point was, essential to applying prospect theory (Fischhoff, 1983). Other strategies included asking people which frame was “most natural” before choosing an option, revising the description to highlight one reference point, changing the order in which frames are presented, changing the outcomes from lives to dollars, and asking which frame people prefer, in case “most natural” evoked a social norm (and not just a frame of reference).

Roughly speaking, whatever I did, about one third of people preferred the certain outcome and one third preferred frame (c), in which that outcome should be preferred, according to prospect theory. However, the two preferences were unrelated at the individual level. That is, people who preferred the 50-lives-lost frame were as likely to choose the certain option as were people who preferred the other two frames. Thus, these manipulations found that, at the population level, one could predict the proportion of people preferring a reference point and the associated prospect theory choice—but not at the individual level.¹² Those predictions may be enough for some purposes (e.g., marketing), while not saying much about individual behavior (Puto, 1987).

5 Archival analysis of representativeness

Not long after *Thinking, Fast and Slow* (Kahneman, 2011) came out, a trauma surgeon came to my office with a copy, saying that it helped explain the seemingly refractory problem of *undertriage*. Whatever the American College of Surgeons and others did, local hospitals did not transfer roughly half of the patients who were injured seriously enough to require treatment at a major medical center. The surgeon, Deepika Mohan, proposed that, when making triage decisions, local doctors rely on the representativeness heuristic, whereby the probability of an event depends on how well it reflects the salient properties of the process that produced it. Patients were undertriaged when the severity of their injuries was not representative of the event that brought them to the local emergency room.

Deepika’s archetypal case was an older person who had fallen, with no visible injuries. Intracranial bleeding was possible, as cause or effect. The local hospital runs a CAT scan to tell. However, if the results are positive, the “golden hours” for benefiting from transfer will have passed. In Deepika’s view, local hospitals have many medically difficult cases, but few diagnostically difficult ones. Because these

¹² Amos, who liked this study, pointed this out. Danny didn’t like it. What a joy it would have been to moderate an adversarial collaboration with them on the topic (Kahneman & Klein, 2009).

physicians' heuristics generally serve them well, the challenge was to improve, not replace, them.

Deepika's diagnosis rang true. However, I was stumped about how to make the auxiliary assumptions needed to define representativeness in that complex situation (Kahneman & Tversky, 1972). The classic demonstrations of representativeness look at properties of the data (e.g., sample size, randomness), rather than properties of the process that produced them. Maya Bar-Hillel and I had once had success in predicting reliance on base rates by asking people to rate how well a person described in a vignette resembled category archetypes (Bar-Hillel & Fischhoff, 1981; Fischhoff & Bar-Hillel, 1984).¹³ However, that form of direct assessment of auxiliary assumption was not possible with physicians.

Instead, we looked at an alternative explanation: local hospitals keep patients whom they should transfer, not because their heuristics fail them, but because they have incentives to keep those patients. Perhaps they want the business. Perhaps they want to demonstrate their prowess. Distinguishing these explanations is a signal detection theory (SDT) problem. A first study, analyzing physicians' judgments of detailed clinical cases in SDT terms, found that some had good discrimination ability and poor decision rules, whereas others had the opposite (Mohan et al., 2012). However, even though the study used unusually realistic stimuli, compared to most psychology studies, medical reviewers complained about the lack of verisimilitude. Ensuing studies involved increasingly intense interactive simulations (e.g., Mohan et al., 2017, 2018) and analyses of electronic medical records (EMRs) (Mohan et al., 2014).

Eventually, the studies laid the foundation for addressing Deepika's original hypothesis (Kulkarni et al., 2019). Clinical experts identified six sources of critical injuries that were common enough to be analyzed in EMRs (e.g., spinal injury, long bone fracture). Based on the preceding studies and clinical experience, the experts identified six characteristics that physicians would see as representative of at least one of the six injury sources and that could also be extracted from EMRs (e.g., neurological deficit, hypoxia). Review of 235,605 patient visits with a primary injury diagnosis found 3199 cases with moderate-to-severe injuries, 24% of whom had been transferred. Patients with at least one representative characteristic were 70% more likely to be transferred than were patients with none.

Local hospitals cannot simply triage more marginal cases, in order to reduce the risk of missing ones that need added care. Medical centers would be overwhelmed. Physicians need to make better judgments. Deepika's strategy, tuned to the intense environment of emergency departments, is to give physicians an intuitive sense for cases that are unrepresentative of their source and then use their medical knowledge to understand why. In Kahneman's terms, she is trying to interrupt System 1 thinking, when it reaches its limits, in order to mobilize System 2. She is having some success (e.g., Mohan et al., 2017, 2018, 2021).

¹³ We used the same vignettes with categories differing in the similarities of their archetypes. We wondered whether people would rely on base rates more when the archetypes were very similar—and representativeness provided less of a guide. Our limiting case had the categories of dentists and orthodontists. Memorably, one participant asked whether the orthodontists were more mellow.

6 Generalization

Kahneman and Tversky's theories have prompted a vast universe of experimental studies testing auxiliary assumptions essential to interpreting the elegant results of the classic studies. Their work has also prompted many speculative extrapolations to real-world settings with far fewer interpretative constraints. "Bias" and, to a lesser extent, "heuristic" have become memes, often invoked in ways far removed from their scientific roots.

For science to progress, it needs systematic ways to evaluate auxiliary assumptions. Three approaches are illustrated above. In these illustrations:

- Studies using direct assessment found measures of availability that correlated with factors predicted to affect frequency judgments that rely on that heuristic. From a construct validity perspective, these results support the measures, the theory of heuristic judgment, and the token account of frequency encoding in memory.
- Studies attempting to manipulate a reference point failed to identify the one that individuals used. From a construct validity perspective, either the manipulations were flawed or there are limits to applying prospect theory to predict individual choices (with some success at the group level).
- Studies analyzing archival data found that trauma triage decisions were consistent with judgment by representativeness. From a construct validity perspective, those results affirm the heuristic account and its interpretation in Medicare records.

The more detailed a task, the clearer the limits to generalizing results. With trauma triage, having a representative characteristic was predictive for some injury sources, but not others. With more general tasks, such as producing examples from memory or selecting reference points, patterns might emerge if results are aggregated systematically across studies. Grether and Plott (1979) pursued that strategy. They created a table with 11 columns, each representing a preference reversal study, and 23 rows, each representing a possible explanation of the results. The cell entries reflect whether each explanation can account for that study's results (E) or not (N). Rows with consistent patterns strengthen confidence in that auxiliary assumption, for whatever universe of choices the studies represent. For example, every cell has N for "confusion and misunderstanding." One possible explanation is that psychologists' disciplinary training, bolstered by peer review, enables them to produce comprehensible tasks—at least for psychologists (Sarah Lichtenstein & Paul Slovic) and tasks (structured gambles) like those in the table.

The more broadly that result is observed, the stronger is the auxiliary assumption that psychologists can be trusted to create sound tasks. Many tests of that assumption are currently underway. Psychology's slowly growing reproducibility crisis was crystalized by a landmark study finding that results failed to replicate for roughly 2/3 of 50 published results in social psychology and 1/3 of 50 in cognitive psychology (Open Science Collaboration, 2015). Many replication studies have followed, testing auxiliary assumptions about which results to trust (Camerer et al., 2018; Hardwicke & Wagenmakers, 2023). These studies have prompted calls for improved research

methods, at first voluntary (e.g., badges for preregistering analyses) and now increasingly required (Hardwicke & Vazier, 2023).

The Cochrane Collaboration’s CONSORT (Consolidated Standards for Reporting Trials) criteria formalize evaluating auxiliary assumptions that determine study quality (Butcher et al., 2022). Those criteria are used to weight clinical trials in meta-analyses. Their terms apply to evaluating evidence in other domains (Fischhoff & Davis, 2014). In some cases, there are enough clinical trials with and without a flaw (e.g., non-blinding to treatment condition) to estimate its impact on treatment effects, allowing predictions of what the results might have been had the study been better. We did that with field trials of home electricity consumptions programs and found that any effect on conservation vanished, after correcting for methodological biases (Davis et al., 2013).¹⁴

Systematic aggregation of results is relatively straightforward for auxiliary assumptions about methodological constructs with clear, shared definitions in different settings, such as measurement quality and investigator training. Substantive auxiliary assumptions, like those regarding memory processes and reference points, require a theoretical context. One row in the Grether and Plott (1979) matrix is “mis-specified incentives.” In this journal, Camerer and Hogarth (1999) reviewed the effects of incentives in 74 experiments. They found no overall effect on judgment task performance, but did find context-dependent effects consistent with a theoretical framework that they proposed, informed by other research. In the journal’s second issue, Lita Furby and I (Fischhoff & Furby, 1988) offered a general framework for auxiliary assumptions in stated preference studies, focused on contingent valuation. Long ago, I offered such a framework for debiasing interventions, reflecting different assumptions about psychological processes (Fischhoff, 1982; Milkman et al., 2009).¹⁵ Other frameworks are certainly possible.¹⁶

7 Orderly foundations for scientific upheaval

In the classic essay introducing Bayesian inference to psychology (Edwards et al., 1963), Ward Edwards, one of Amos’ mentors, touted the value of the “interocular traumatic effect,” so strong that it hits you between the eyes. Ward was referring to evidence that, for practical purposes, led to accepting one of a set of competing hypotheses.¹⁷ Many studies seek such evidence, attempting to resolve, once and for all, say, whether frequency information is stored in memory as types or tokens. That aspiration requires advocates of competing hypotheses to agree on the interpretation of evidence that might distinguish them.¹⁸ Danny Kahneman’s advocacy of adver-

¹⁴ A related study found a Hawthorne effect on home electricity consumption equal to the effect of elaborate interventions, testing the auxiliary assumption that knowing that they are in a study does not affect participants’ behavior (Schwartz et al., 2013).

¹⁵ For more on the robustness of hindsight bias, see Fischhoff (in press).

¹⁶ For an early career attempt to scope them all, see Fischhoff et al. (1980).

¹⁷ Respecting the effective convergence principle in Bayesian inference.

¹⁸ Lakatos’ account of the myth of the critical experiment is relevant here (Lakatos, 1970).

sarial collaboration promoted such studies, which have greater chance of success when the parties share auxiliary assumptions allowing them to isolate disagreements that jointly designed studies might resolve (e.g., Kahneman & Klein, 2009).

Scientists sometimes propose accounts that violate the tenets of Bayesian inference by introducing hypotheses that were not previously considered, hence lack priors that can be updated. Successful proposals reinterpret existing evidence, while offering novel evidence, whose collection they motivate. They must be close enough to previous research to be interpretable, given shared auxiliary assumptions, but far enough to shake faith in some of those assumptions.¹⁹

In science, such innovations are rare. Producing two in a decade, as Tversky and Kahneman did, is truly remarkable. Their innovations created a kind of manageable chaos, wherein scientists could apply their conventional tools to novel questions. That success reflected not only Danny and Amos' immense creativity, but also their mastery of the auxiliary assumption conventions of their respective branches of psychology. As decision science has grown in size and complexity, such grand syntheses have become increasingly difficult.²⁰ There are too many auxiliary assumptions to consider, especially with tasks from diverse disciplines, each with its interpretative norms. In the Grether and Plott matrix, for example, some cell entries were important to psychologists, but irrelevant to economists.

Systematic reviews of evidence regarding auxiliary assumptions can help signals emerge from the noisy chaos of vibrant science. Deciding where studies go in a matrix can refine thinking about the assumptions. Once aggregated, the evidence may reveal topics that can be put to rest and topics that have been neglected. For example, a review of prejudice-reduction interventions found that almost all assumed that changes were stable, leading researchers to measure attitudes immediately before and after interventions (Lai et al., 2016). A review of the 3.7% of studies that measured "after" attitudes on a following day found no correlation between individuals' before and after attitudes (Vuletic & Payne, 2019). Another review found that almost all studies made the auxiliary assumption that "light touch" (one-shot) interventions provided true measures of prejudice malleability (Paluck et al., 2021).

The Cochrane Collaboration's CONSORT criteria and the Open Science Framework provide institutional homes for systematic reviews focused on methodological assumptions, refining the set of studies worth considering.²¹ Two National Institutes of Health (NIH) initiatives created frameworks for evaluating substantive assumptions, refining the set of measures worth considering. One, proposed by the NIH Behavior Change Consortium, specified 16 features of sound *treatment fidelity* that funders, editors, and reviewers should assess when evaluating interventions (Belig et al., 2004).²²

¹⁹ What Lakatos (1970) might have called sophisticated falsificationism, in his reflection on scientific revolutions.

²⁰ Ward Edwards may have realized that as he moved on from the hope for "A theory of decision making," as expressed in the title of his seminal article (Edwards, 1954).

²¹ Cochrane has sometimes struggled in extending from medical interventions to behavioral ones (Soares-Weiser et al., 2020).

²² One critique argued that without accepted measures for most of those features, the hoped-for standardization is notional (Leventhal & Friedman, 2004). That critique also questioned whether one framework

The second initiative created a framework for assessing patient quality of life. A proliferation of measures (e.g., for pain) had complicated comparing results across studies. NIH supported creating the PROMIS[®] (Patient-Reported Outcome Measurement Information System) battery of psychometrically sound measures (Cella et al., 2007). NIH does not require funded studies to use the PROMIS measures. Rather, it provides the incentives of quality assurance, comparability, and cost savings.²³ There are currently PROMIS measures for 44 quality-of-life outcomes, some with utility functions, based on standard gambles (Dewitt et al., 2018; although see Dewitt et al., 2019).

8 Conclusion

Feyerabend (1975) argued that any attempt to mandate practices violated the tenets of science. Kahneman and Tversky's work was the kind of irreverent breakthrough that he might have appreciated, challenging conventional auxiliary assumptions. With judgment by heuristics, one such assumption was that the primitives of thought could be mapped onto those of Bayesian probability theory; a second was that epistemic heuristics had a place alongside deterministic ones (e.g., Newell & Simon, 1972). With prospect theory, one challenged assumption was that the primitives of thought could be mapped onto those of decision theory; a second was that a structured model could capture thought processes and not just input-output relationships (Dawes, 1979).

These auxiliary assumptions were clear enough that Tversky and Kahneman, given their insight and determination, could see their limits and create alternatives that other scientists found useful. Like any breakthrough, theirs required new auxiliary assumptions. Examining them has proven empirically and theoretically productive. Grether and Plott (1979) gave an early inkling of that promise and helped structure early interactions between psychology and economics. Subsequent exchanges have often been productive, if occasionally contentious, facilitated by shared primitives (probability and utility theory) and by partial agreement on the auxiliary assumptions of empirical research (Bruine de Bruin & Fischhoff, 2017; Manski, 2017).

When systematic reviews find consistent patterns with auxiliary assumptions, the science can move on with greater confidence. For situations where that evidence is lacking, this article illustrates three approaches to creating it. In the spirit of the collaborative science reflected in Danny's advocacy of adversarial collaboration, systematically evaluating auxiliary assumptions can clarify meaningful disagreements, increasing the chances of mutually accepted results and the rare breakthrough, posing new auxiliary assumptions.

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could suit all conditions and interventions.

²³ PROMIS measures are free, unlike proprietary ones. They use item-response theory to produce efficient sets of items.

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Declarations

Competing interests I declare no competing interests.

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